# SPLITTING OF SPHERICAL-TWIST SYMPLECTIC MONODROMY FOR CALABI-YAU THREEFOLDS FROM MARKED BORDERED SURFACES

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*Allowed standard prerequisites (editorial).*

- Autoequivalence groups and connected stability spaces of type A Calabi-Yau categories. Ikeda Theorems7.15,7.16; Qiu-Woolf Theorems B; BS Theorem9.9, all cited precisely in author proof
- Cluster categories and exchange-graph quotient. Baur-Marsh, BRT, King-QiuYu-cluster Theorem8.6
- Koszul duality and Rabinowitz/cluster equivalence for Dynkin plumbings. AbSm-plumbing, Kalck-Yang, BJK Corollary1.4 and Lekili-Ueda as explicitly cited

*Source reference.* Original section 7: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 1307–1307. Forward pointer in introductory categorical scope only, unrelated to the type A splitting proof.

*Source reference.* Original theorem 3.4: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 474–476. Source narrative comparison to another independent geometric case only; none of its results is used as an input to the retained proof.

*Source reference.* Original proposition 3.5: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 480–482. Source narrative comparison to another independent geometric case only; none of its results is used as an input to the retained proof.

*Source reference.* Original reference 3: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 657–664. Source narrative comparison to another independent geometric case only; none of its results is used as an input to the retained proof.

## 1. Flavours of Fukaya category

Let $(X, \omega)$ be a connected symplectic manifold with $2c_1(X) = 0$. We will fix a Maslov cover of the Lagrangian Grassmannian, meaning that there is a well-defined group $\mathrm{Symp}^{\mathrm{gr}}(X)$ of graded symplectomorphisms, and a subgroup $\mathbb{Z} \leqslant \mathrm{Symp}^{\mathrm{gr}}(X)$ generated by the shift (i.e., the fibrewise deck transformation). Recall from [74, Lemma 2.4] the following.

**Lemma 1.1.** *There is an exact sequence*

$$1 \to \mathbb{Z} \to \mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X) \to H^1(X; \mathbb{R}).$$

*The final map is not a homomorphism but its kernel is a subgroup.*

**Lemma 1.2.** *Suppose that $X$ is non-compact. The forgetful map $\mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X)$ is split over the subgroup $\mathrm{Symp}_{\mathrm{ct}}(X)$.*

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*Proof.* Take the unique gradings for symplectomorphisms which act trivially on the fibres of the Maslov cover near infinity. □

Let $(X, \omega)$ be a Liouville manifold, with contact type boundary, and assume $2c_1(X) = 0$. We will be interested in all of

(1) the compact Fukaya category $\mathcal{F}(X)$, with objects compact graded exact spin Lagrangians;
(2) the wrapped Fukaya category $\mathcal{W}(X)$, with objects graded exact spin Lagrangians which are cones over Legendrian submanifolds near infinity;
(3) the Rabinowitz Fukaya category $\mathcal{R}(X) := \mathcal{W}(X)/\mathcal{W}(X)^{\mathrm{prop}}$, which is the Verdier quotient of the (Yoneda image of the) wrapped category by its subcategory of proper modules.

All three categories are defined over $\mathbb{Z}$, and $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a full and faithful subcategory. We say that 'Koszul duality' holds for the compact and wrapped categories if $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a Koszul dualising subcategory in the sense of [26, Definition 2.12]. This often holds for Weinstein manifolds obtained from plumbings of simply connected pieces. Concretely, the wrapped category of a Weinstein manifold admits a finite set of generators $\{T_i \mid 1 \leqslant i \leqslant N\}$ and one seeks a finite set of compact branes $\{L_i \mid 1 \leqslant i \leqslant N\}$ such that the total endomorphism algebras $\mathcal{Q} = \bigoplus_{i,j} HF^*(L_i, L_j)$ and $\mathcal{A} = \bigoplus_{i,j} HW^*(T_i, T_j)$, which are augmented over the semisimple ring $\Bbbk = \bigoplus_{i=1}^N \mathbb{K}_i$ generated by the units of the objects, are $A_\infty$-Koszul dual, meaning

$$\mathcal{Q} \simeq \mathrm{Ext}_{\mathcal{A}}(\Bbbk, \Bbbk), \qquad \mathcal{A} \simeq \mathrm{Ext}_{\mathcal{Q}}(\Bbbk, \Bbbk).$$

In this setting, there is an equivalence [26, Corollary 1.4]

$$\mathcal{R}(X) \simeq \mathcal{W}(X)/\mathcal{F}(X).$$

This is the formulation we will encounter in practise, but the general algebraic construction of $\mathcal{R}(X)$ from $\mathcal{W}(X)$ in particular shows that autoequivalences of the latter act on the former. (In Section 7, it will be important to work over $\mathbb{C}$, to incorporate $\mathbb{C}^*$-valued local systems on Lagrangians.)

Any autoequivalence of $\mathcal{W}(X)$ arising from a symplectomorphism of $X$ will preserve the compact category $\mathcal{F}(X)$, and hence induce an equivalence of $\mathcal{R}(X)$. For background on $\mathcal{F}(X)$, see [76], and for background on $\mathcal{W}(X)$, see [1]; for background on derived localisation in general, see [22], and in the context of $\mathcal{R}(X)$ specifically, see [26].

The Grothendieck group $K(\mathcal{F}(X))$ is the free abelian group on isomorphism classes of objects, modulo relations coming from exact triangles. For later use, we define the 'numerical Grothendieck group'

$$K_{\mathrm{num}}(\mathcal{F}(X)) \tag{1.1}$$

to be the quotient of $K(\mathcal{F}(X))$ by the subgroup which is the radical of the Euler pairing, i.e., that generated by compact exact Lagrangians $L \in \mathrm{Ob}\,\mathcal{F}(X)$ for which $\chi(HF^*(L, \bullet)) = 0$ for every $\bullet \in \mathcal{W}(X)$.

We will be interested in actions of symplectomorphism groups on Fukaya categories. Especially for actions on the wrapped category, one needs some control of maps at infinity to ensure that (perhaps after isotopy) they preserve the class of Lagrangians which are conical[1] at infinity. We say a symplectomorphism $\phi$ of a Liouville manifold $(X, \mathrm{d}\theta)$ is *exact* if $\phi^*\theta = \theta + \mathrm{d}f$ for some $f\colon X \to \mathbb{R}$, and *strongly exact* if one can take $f$ to have compact support. Both notions have obvious graded

[1]Or more generally geometrically bounded for a dissipative almost complex structure in the sense of Groman [29].

analogues. It is classical that strongly exact maps act on the wrapped category. Write $\mathrm{Symp}_{\mathrm{ex}}(X)$ for the group of exact symplectomorphisms.

**Lemma 1.3.** *Assume that the Liouville manifold $X$ has finite type, so can be written as the completion of a Liouville domain $\mathring{X}$. Then*

(1) *There is a homomorphism $\pi_0 \mathrm{Symp}_{\mathrm{ct}}(X) \to \pi_0 \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$;*
(2) *The group $\pi_0 \mathrm{Symp}(\mathring{X})$ acts on both $\mathcal{F}(X)$ and $\mathcal{W}(X)$.*

*Proof.* For the first statement, conjugation by a suitably large Liouville flow will bring the support into $\mathring{X}$. We claim we can then deform to obtain an exact representative of the map. Let $\omega = \mathrm{d}\lambda$ be the Liouville form on $X$, and suppose we are given $\psi \in \mathrm{Symp}_{\mathrm{ct}}(X)$. We have $\psi^*\lambda - \lambda = \theta$, where $\theta$ is a closed one-form with compact support. Let $V$ be the vector field which is symplectically dual to $-\theta$. This has compact support, and so is integrable for all time; let $\sigma_t$ denote its flow. Then applying Cartan's formula, we get that $\psi \circ \sigma_1$ is an exact symplectomorphism. (Note that the proof of [11, Lemma 1.1] specialises to this argument in this case.)

This construction can be performed in families, which shows that the inclusion

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \hookrightarrow \mathrm{Symp}_{\mathrm{ct}}(X)$$

is a homotopy equivalence, so induces an isomorphism on $\pi_0$. On the other hand, there is then a forgetful map

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \to \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$$

which induces a homomorphism on $\pi_0$. Combining these two gives the first statement.

The second statement is exactly [45, Corollary 2.4] (cf. [46], which in particular involves deforming a map of the Liouville domain to be exact, using a version of the argument given in the previous paragraphs). □

**Lemma 1.4.** *In the setting of Lemma* 1.3*, assume moreover that $2c_1(\mathring{X}) = 0$. Then there are compatible representations*

$$\begin{array}{ccc} \pi_0 \mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X) & \longrightarrow & \mathrm{Auteq}(\mathcal{F}(X)) \\ \downarrow & & \uparrow \\ \pi_0 \mathrm{Symp}^{\mathrm{gr}}(\mathring{X}) & \longrightarrow & \mathrm{Auteq}(\mathcal{W}(X)). \end{array}$$

The top line is classical; the bottom line is [45, Corollary 2.9] (see also [46]).

Although the Rabinowitz category is not a priori determined by the contact boundary of $X$, one still has the following lemma.

**Lemma 1.5.** *The group $\pi_0 \mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X)$ acts trivially on the Rabinowitz category.*

*Proof.* Let $L$, $K$ be Lagrangians which are cones over Legendrian submanifolds $L_\infty$ and $K_\infty$. The morphism group $RFC^*(L,K)$ in $\mathcal{R}(X)$ between $L$ and $K$ is the chain-level mapping cone of a natural continuation map $CW_*(L,K) \to CW^*(L,K)$ coming from a monotone homotopy of Hamiltonians from 'infinite negative wrapping to infinite positive wrapping'. This factors as a composition

$$CW_*(L,K) \xrightarrow[a]{} CF^*(L, \phi_{-\varepsilon}K) \xrightarrow[c]{} CF^*(L, \phi_{+\varepsilon}K) \xrightarrow[b]{} CW^*(L,K),$$

where $a$, $b$ are projection respectively inclusion maps of summands into the relevant direct systems and $c$ is the natural continuation from small negative to small positive wrapping. Viewing $RFC^*(L,K) = \mathrm{Cone}(b \circ c \circ a)$, there are exact triangles of mapping cones

$$\mathrm{Cone}(c \circ a) \to RFC^*(L,K) \to \mathrm{Cone}(b)$$

and

$$\mathrm{Cone}(a) \to \mathrm{Cone}(c \circ a) \to \mathrm{Cone}(c).$$

The continuation map $c$ induces the Poincaré duality isomorphism in Lagrangian Floer theory

$$HF^*(L, \phi_{-\varepsilon} K) \to HF^*(L, \phi_{+\varepsilon}(K))$$

(which recovers the isomorphism $H_*(L, \partial_\infty L) \to H^{n-*}(L)$ when $L = K$), so has acyclic cone. On the other hand, $\mathrm{Cone}(b) \simeq CW(L,K)^+$ defines the Lagrangian Floer analogue of 'positive symplectic cohomology' [70]; it can be computed by an action truncated version of the wrapped Floer complex which removes action zero chords (i.e., interior intersection points), yielding a complex only built from Reeb chords at infinity between the Legendrian boundaries. Similarly for $\mathrm{Cone}(a)$. Putting these ingredients together, one obtains a cochain complex quasi-isomorphic to that underlying $RFC^*(L,K)$ and which is generated only by Reeb chords at the boundary. Any compactly supported map then acts trivially on the underlying chain group.

Since $L$ and $K$ were arbitrary, compactly supported maps act trivially on the whole $A_\infty$-category. □

To conclude this section, we briefly recall the categories associated to quivers with potential; a more extensive discussion can be found in [83].

A quiver $Q$ is a directed graph, and a potential $W$ on $Q$ is a formal sum of cycles in the edges. To such $(Q,W)$ one can associate a 3-dimensional[2] Calabi–Yau category $\mathcal{C}(Q,W)$ via the 'Ginzburg construction' [28], which defines a non-positively graded *dg*-algebra $\mathcal{J}(Q,W)$ with $H^0(\mathcal{J}(Q,W)) = J(Q,W)$ the Jacobi algebra (i.e., the path algebra modulo cyclic derivatives of the potential). We always assume that $W$ has no quadratic terms; then the category $\mathcal{C}$ has a spherical object for each vertex of $Q$, and $W$ encodes the non-trivial products in a cyclic $A_\infty$-structure on the total endomorphism algebra of those objects (this is the Koszul dual of the Ginzburg algebra). By a 'symplectic geometric model' for $\mathcal{C}(Q,W)$ we mean a configuration of Lagrangian spheres $\{L_v\}_{v \in \mathrm{Vert}(Q)} \subset (X, \omega)$ such that the number of edges between $v$ and $v'$ gives the rank of $HF^*(L_v, L_{v'})$ and the endomorphism algebra $\bigoplus_{v,v'} HF^*(L_v, L_{v'})$ is $A_\infty$-quasi-isomorphic to that determined by the potential $W$. Such models are especially useful when the $\{L_v\}$ (split-)generate the compact category $\mathcal{F}(X,\omega)$. One can also seek algebro-geometric models for $\mathcal{C}(Q,W)$, i.e., embeddings $\mathcal{C}(Q,W) \hookrightarrow \mathcal{D}(Y)$ into the derived category of some quasi-projective $Y$ (ideally with image some naturally characterised subcategory), or indeed algebraic models in terms of realisations in derived categories of modules over a non-commutative algebra. All of these will play a role in the sequel.

## 2. Quivers and threefolds from marked bordered surfaces

The type $A$ Milnor fibres $A_k^{(n)}$ have a quite different generalisation from that of Question 3, namely affine Calabi–Yau varieties which are Lefschetz fibrations with general fibre $T^*S^{n-1}$ over a non-compact surface. We will focus on the case of 3-folds fibred in $T^*S^2$. In this case, the space of stability conditions is not known to be connected, but we can access an analogue of Proposition 3.5 and Theorem 3.4 using knowledge of spherical objects. (Note that in more complicated cases the relevant braid-type group may have no center, in which case these two results are essentially equivalent.)

Let $S$ be an oriented surface of genus $g$ with a non-empty set of $d$ marked points $\{p_1, \ldots, p_d\}$. Fix an integer $d_i > 2$ for each $p_i$; we encode this information by a vector $\mathbf{d}$. We take the real blow-up of $S$ at the $\{p_i\}$ and mark $d_i - 2$ points

[2]When the arrows admit suitable gradings, one can associate a $\mathrm{CY}_n$-category for any $n$.

on the $i$-th boundary component; let $M$ denote the resulting collection of marked points. We consider ideal triangulations of $S$ in the sense of [25], i.e., maximal collections of disjoint arcs, where an arc is a smooth embedded path connecting points of $M \subset \partial S$, with interior in $S \backslash \partial S$. (To get something visibly resembling a triangulation one should include arcs in $\partial S$ joining points of $M$, but these are *not* included in the set of arcs.) A triangle is 'interior' if its closure meets $\partial S$ only in $M$. Any ideal triangulation has

$$6g - 6 + \sum_i (d_i + 1) \text{ edges} \qquad \text{and} \qquad 4g - 4 + \sum_i d_i \text{ faces} \tag{2.1}$$

(we are in the case with 'no punctures' in the terminology of [12], and we are not including the boundary arcs between points of $M$ as edges).

We will be interested in the cellulation which is dual to the given ideal triangulation, whose vertices will correspond to simple zeroes of a quadratic differential below.

There is a quiver associated to the ideal triangulation, with vertices the midpoints of the edges and an arrow when two open arcs share a vertex (in the direction in which one is anticlockwise from the other). See Figure 1 for an illustration[3] (in a region interior to the surface, so $\partial S$ is not visible in the figure). The resulting collection of arrows in particular inscribes a clockwise 3-cycle in each interior triangle, so comes with an obvious collection of 3-cycles: the triangles just inscribed. This defines a canonical potential

$$W = \sum (\circlearrowright \text{triangles})$$

and hence via the Ginzburg construction (cf. the brief recollection at the end of Section 1) a $\mathrm{CY}_3$-category $\mathcal{C}(Q, W)$. One can show this category depends only on the pair $(S, M)$ or equivalently on $(g, \mathbf{d})$. More precisely, flips of the triangulation induce mutations of quivers-with-potential, which yield quasi-equivalent categories by [48]. We may write $\mathcal{C}(S, M)$ or $\mathcal{C}(g, \mathbf{d})$ for a category in this equivalence class.

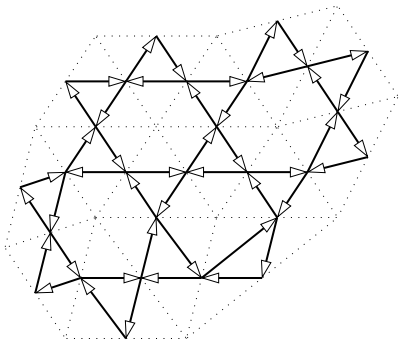


FIGURE 1. Quiver inscribed in a triangulation.

Let $\phi$ be a meromorphic quadratic differential on a surface $S$ with poles of order $d_i$ at $p_i$ and with distinct zeroes. Let $D \subset S$ be the set of poles. Let $Y = Y_\phi \to S \backslash D$ be the Lefschetz fibration with smooth fibre $T^*S^2$ and with nodal singular fibres over the zeroes of $\phi$. This is a non-compact affine Calabi–Yau 3-fold, which moves in a family over the complex orbifold $\mathrm{Quad}(g, \mathbf{d})$ of quadratic differentials with the prescribed pole orders and with simple zeroes. An embedded 'matching path' in $S$ between zeroes of $\phi$ defines a Lagrangian 3-sphere in $Y_\phi$.

[3]This figure is borrowed from [12].

**Lemma 2.1.** *An ideal triangulation defines a collection of matching paths, which form the edges of the dual cellulation.*

See [83] for a proof, and further background.

Let $\Gamma(S, M)$ denote the mapping class group of the associated marked bordered surface.[4] Let $\mathrm{Sph}(S, M)$ denote the group generated by spherical twists in matching paths dual to the edges of the ideal triangulation.

**Theorem 2.2** ([12, 83])**.** *There is a distinguished connected component* $\mathrm{Stab}^\dagger(\mathcal{C}(S, M))$ *of the space of stability conditions on* $\mathcal{C}(S, M)$*, and a subgroup* $\mathrm{Auteq}^\dagger(\mathcal{C}) \subset \mathrm{Auteq}(\mathcal{C})$ *of autoequivalences preserving that component, with the following properties:*

- $\mathrm{Stab}^\dagger(\mathcal{C}(S, M))/\mathrm{Auteq}^\dagger(\mathcal{C}) \cong \mathrm{Quad}(g, \mathbf{d})$ *is the complex orbifold whose points parametrise biholomorphism classes of pairs of a Riemann surface* $S$ *of genus* $g$ *and a meromorphic quadratic differential on* $S$ *with poles of orders* $d_i > 2$ *and with simple zeroes.*
- *There is an exact sequence*

$$1 \to \mathrm{Sph}(S, M) \to \mathrm{Auteq}^\dagger(\mathcal{C}(S, M)) \to \Gamma(S, M) \to 1.$$

- *There is an embedding* $\mathcal{C}(Q, W) \hookrightarrow \mathcal{F}(Y_\phi)$ *for* $(Q, W)$ *the quiver with potential associated to any ideal triangulation of* $(S, M)$*.*

**Remark 2.3.** Let $Z \subset \mathrm{interior}(S) \subset S \backslash M$ denote a set of points in bijection with the (assumed simple) zeroes of a quadratic differential $\phi$ of the prescribed polar type. We can consider the mapping class group $\Gamma(S, M; Z)$ of mapping classes which fix $M$ setwise and which preserve the finite set $Z$ setwise. This fits into a short exact sequence

$$1 \to \mathrm{SBr}(S, Z) \to \Gamma(S, M; Z) \to \Gamma(S, M) \to 1,$$

where the first term is the 'surface braid group' of $(S, Z)$, i.e., the fundamental group of the space of configurations of unordered $|Z|$-tuples of points on $S$. The surface braid group has two kinds of generators: braid-type maps associated to a matching path between two distinct points of $Z$, and point-pushing maps associated to a non-trivial loop on $S$ and a single point of $Z$. The paper [66] has shown that $\mathrm{Sph}(S, M)$ is the subgroup of $\mathrm{SBr}(S, Z)$ generated by the first type of map.

Let $\Sigma \to S$ be the spectral curve, which is the double cover branched at $Z \subset S$, and let $\iota \colon \Sigma \to \Sigma$ denote the covering involution. Let $\Gamma(\Sigma, \iota)$ denote the hyperelliptic-type mapping class group of isotopy classes of diffeomorphisms of $\Sigma$ which commute with $\iota$. Associating to a braid-twist in a matching path $\gamma$ on $S$ with end-points in $Z$ the Dehn twist in the $\iota$-invariant simple closed curve $\hat{\gamma} \subset \Sigma$ which is the lift of $\gamma$ shows that there is an embedding

$$\mathrm{Sph}(S, M) \hookrightarrow \Gamma(\Sigma, \iota).$$

According to the work of [9] on symmetric mapping class groups, cf. also [61], and noting that we might as well restrict to the part of the autoequivalence group which maps to $\Gamma(S, M_=)$ which fixes $M$ and hence $\partial S$ pointwise, we find that when $\chi(S) < 0$, then $\mathrm{Sph}(S, M)$ is characterised as the centraliser of the involution $\iota$ in $\Gamma(\Sigma, \partial\Sigma)$. (We do not need to divide by $\iota$ since the mapping class group of a surface with non-empty boundary, fixing the boundary pointwise, is torsion-free, so $\iota$ does not define an element of $\Gamma(\Sigma, \partial\Sigma)$.)

[4]For comparison to [12, 83], note that we are in the case of an unpunctured surface with non-empty boundary.

The group $\mathrm{Sph}(S, M)$ acts via spherical twists which are geometrically realised by Dehn twists in Lagrangian matching spheres, so there is a natural representation

$$\mathrm{Sph}(S, M) \to \pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y).$$

The following generalises (the 3-dimensional instance of) Proposition 3.5.

**Theorem 2.4.** *The monodromy map* $\rho\colon \mathrm{Sph}(S, M) \to \pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y)$ *splits.*

Recall that a spherical object is reachable if it is one of the simples in a reachable heart, i.e., one obtained from the standard heart by a finite sequence of tilts.

**Lemma 2.5.** *The set of reachable spherical objects in* $\mathcal{C}(Q, W)$ *is in bijection with the set of Lagrangian matching spheres.*

*Proof.* This is proved in [51] (note we are in the 'unpunctured case' which they treat). □

**Lemma 2.6.** *Every Lagrangian* 3*-sphere in* $Y_\phi$ *is quasi-isomorphic to the Lagrangian* 3*-sphere associated to some* (*unique up to isotopy*) *embedded matching path.*

*Proof.* Let $L \subset Y_\phi$ be a Lagrangian 3-sphere. Then $L$ lifts to a Lagrangian $\tilde{L}$ in the universal cover $\pi\colon \tilde{Y} \to Y$, which is a Lefschetz fibration $\tilde{Y} \to \mathbb{C}$ with fibre $T^*S^2$ and infinitely many Lefschetz singular fibres. This is an 'infinite' Milnor fibre of type $A$ in dimension 3. Being compact, any lift of $L$ lives in a finite subspace which is the total space of a fibration $A_k^{(3)}$ for some $k \gg 0$. Spherical objects in $A_k^{(3)}$ are classified; for instance, this follows from the result of [51] together with the fact that the heart exchange graph is connected in type $A$ [38], which means that all hearts are reachable. It follows that there is a unique embedded matching path $\gamma$, with associated Lagrangian sphere $\tilde{L}_\gamma$, such that $\tilde{L}$ is quasi-isomorphic to $\tilde{L}_\gamma$ in the Fukaya category of $A_k^{(3)}$.

Let $\sigma$ denote the immersed matching path downstairs which is the image of $\gamma$, and let $L_\sigma$ be the corresponding immersed matching sphere, which is the image of $\tilde{L}_\gamma$ under $\pi$. We want to show that $L$ and $L_\sigma$ are quasi-isomorphic and that $\sigma$ is embedded. In order to do this, we will construct a quasi-isomorphism upstairs and descend it.

Let $\tilde{L}'_\gamma$ be a small Hamiltonian perturbation of $\tilde{L}_\gamma$. By a small perturbation of $L$ pulled back to $\tilde{L}$, we can assume that the pairs $(\tilde{L}, \tilde{L}_\gamma)$ and $(\tilde{L}, \tilde{L}'_\gamma)$ both intersect transversally. Let $L'_\sigma$ be the image under $\pi$ of $\tilde{L}'_\gamma$. Note that $\gamma$ being embedded means that $\sigma$ bounds no teardrops disjoint from the marked points, which implies the associated immersed (cleanly self-intersecting) Lagrangian 3-sphere $L_\sigma \subset Y_\phi$ is unobstructed. Indeed, the obstruction term $\mathfrak{m}_0$ in immersed Floer theory arises from holomorphic discs with a unique boundary puncture; exactness of the immersion means that this puncture must map to one of the transverse self-intersections, defining a teardrop disc, cf. [3, Section 13.3] for a discussion of related examples.

Pick a regular $J$ on $Y_\phi$ and pull it back to $\tilde{J}$ on $\tilde{Y}$. Since $\tilde{Y} \to Y$ is an unramified cover, and the domain of a holomorphic disc $u\colon D \to Y$ is simply-connected, it admits a unique lift to $\tilde{Y}$ given any fixed point in the preimage, and since the almost complex structure on $\tilde{Y}$ is pulled back from $Y$, this lift is again holomorphic. We will apply this where $D$ is a holomorphic strip or triangle in $Y$, and the point lift which fixes a preimage is an intersection point (using that the map extends over the boundary punctures). This lifting property of holomorphic strips therefore means that the projection map $\pi$ defines a chain map

$$CF_{\tilde{Y}}\big(\tilde{L}_\gamma, \tilde{L}; \tilde{J}\big) \to CF_Y(L_\sigma, L; J) \tag{2.2}$$

and similarly for $\big(\tilde{L}, \tilde{L}_\gamma\big)$ and $\big(\tilde{L}_\gamma, \tilde{L}'_\gamma\big)$.

Pick cycles $\alpha \in CF_{\tilde{Y}}\big(\tilde{L}, \tilde{L}'_\gamma\big)$, $\beta \in CF_{\tilde{Y}}\big(\tilde{L}_\gamma, \tilde{L}\big)$ such that

$$\mu_2(\alpha, \beta) = e \in CF_{\tilde{Y}}\big(\tilde{L}_\gamma, \tilde{L}'_\gamma\big),$$

where $e$ represents the cohomological unit. The images of $\alpha$, $\beta$ and $e$ under $\pi$ all represent cycles by (2.2), say $\pi(\alpha)$, $\pi(\beta)$ and $\pi(e)$. By the same unique lifting for based holomorphic triangles, there is a non-trivial Floer product

$$\mu_2(\pi(\alpha), \pi(\beta)) = \pi(e) \in CF_Y(L_\sigma, L'_\sigma)$$

and $\pi(e)$ again represents the cohomological unit.

Taking the product in the other order (and using $L$, $L_\sigma$ and a small Hamiltonian perturbation of $L$), we deduce that $L$ and $L_\sigma$ are quasi-isomorphic.

Finally, we can assume that $\sigma$ self-intersects minimally. There is a fibrewise involution on the Lefschetz fibration $A_k^{(3)}$, with fixed locus a two-dimensional Riemann surface of finite type, which descends to a fibrewise involution on a compact subdomain of $Y_\phi$ containing $L_\sigma$. The matching sphere $\tilde{L}_\gamma$, and hence its image $L_\sigma$, can be constructed to be setwise preserved by this involution. If $\sigma$ is not embedded, then by the arguments of [49, Section 5] (working with a $\mathbb{Z}/2$-equivariant $J$ and achieving equivariant transversality since there are no curves in the fixed locus for topological reasons), $HF(L_\sigma, L_\sigma; \mathbb{Z}/2)$ has rank $> 2$, which is a contradiction since $L_\sigma$ is spherical. □

**Corollary 2.7.** *$\mathcal{C}(Q, W)$ is a symplectomorphism-invariant subcategory of $\mathcal{F}(Y_\phi)$, and all Lagrangian spheres are reachable spherical objects.*

We now know that $\pi_0 \operatorname{Symp}_{\mathrm{ct}}$ preserves reachable spherical objects, but not at this point reachable hearts.

**Lemma 2.8.** *Let $L$ and $L'$ be two Lagrangian spheres in $Y_\phi$. Then $HF^*(L, L')$ has rank one exactly when the two corresponding matching paths share one end-point.*

*Proof.* Let $\gamma$ and $\gamma'$ denote the relevant matching paths. By isotopy of these (which induces a Hamiltonian isotopy of the corresponding matching Lagrangian 3-spheres), we can assume that the two paths meet minimally. If two sub-arcs of $\gamma \cup \gamma'$ bound a loop which is non-trivial in $\pi_1(S)$ then there can clearly be no Floer strips with those sub-arcs as boundary. Suppose two sub-arcs of the paths bound a region which contains a Lefschetz critical point of the fibration. Then there are no holomorphic strips which project to that region; this follows from the same vanishing result that underlies the proof of the long exact triangle in Floer theory (cf. [76, Lemma 17.4, Remark 17.8] and the argument 'bubbling off' a standard Lefschetz fibration in [76, Section 17d]). Given this, the result is straightforward, and closely modelled on [49] as in the final part of the proof of Lemma 2.6. □

**Lemma 2.9.** *Let $L$ and $L'$ be two Lagrangian spheres in $Y_\phi$ which are not quasi-isomorphic. Then $HF^*(L, L')$ has rank two exactly when*

1. *either the two associated matching paths share both endpoints and are otherwise disjoint;*
2. *or the two associated matching paths meet transversely at exactly one interior point.*

*Proof.* Since $L \not\sim L'$ the associated matching paths $\gamma$ and $\gamma'$ represent different relative homotopy classes in $\pi_1(S, \text{Zeroes})$. If they meet only at both end-points, then since the paths are not homotopic their union traces a non-trivial loop, which means that the union $L_\gamma \cup L_{\gamma'}$ bounds no rigid holomorphic strips, so $HF(L, L')$ is indeed rank two. If they meet transversely at an interior point, then $L_\gamma$ and $L_{\gamma'}$ meet cleanly along an $S^2$ lying above that point. Putting the matching paths in minimal position and arguing as in the proof of Lemma 2.8 shows that any such

$S^2$-component of the intersection locus contributes a two-dimensional vector space to Floer cohomology. □

**Lemma 2.10.** *Suppose $L$, $L'$ are not quasi-isomorphic and $HF^*(L, L')$ has rank two. The two cases of Lemma* 2.9 *cannot be related by a global symplectomorphism.*

*Proof.* Assume first that our differentials $\phi$ have at least five zeroes. We claim that the two cases are distinguished by whether or not there is a third Lagrangian sphere $L''$ with $HF^*(L, L'')$ of rank one and $HF^*(L', L'')$ of rank zero. To see this, suppose $p \in \gamma$ is an end-point of $\gamma$ and $p \notin \gamma'$. Then we are necessarily in the second case, and the union $\gamma \cup \gamma'$ lies in a disc in $S$ and in particular does not separate. Then one can take a third matching path $\sigma$ which meets $\gamma$ at $p$ but is disjoint from $\gamma'$. Suppose then $\phi$ has exactly four zeroes (so the original ideal triangulation had four faces). By (2.1), $S$ is a disc, annulus or a torus with one boundary component. The first two cases correspond to the $A_4$ or affine $A_4$ spaces, where the result is well-known (cf. the proof of Lemma 2.8). In the final case, either the two matching paths sweep a non-trivial loop in $\pi_1(T^2)$, and the result is straightforward, or the whole configuration lies in a disc or annulus, reducing one to a previously considered case.

This shows that a suitable $L''$ exists in case (2) of Lemma 2.9. We claim it cannot exist in case (1). Indeed, if $HF^*(L, L'')$ has rank one, then by Lemma 2.8 $L''$ is associated to a matching path which shares exactly one end-point in common with $L$ and is otherwise disjoint from it. But then $L''$ also shares exactly one end-point with the matching path defining $L'$. This means that (after perturbing to achieve transversality) the rank of $CF(L', L'')$ is odd, since any interior intersection points of the corresponding matching paths contribute two generators from Morsifying an $S^2$-clean intersection. But then $HF(L', L'') \neq 0$. □

The set $\mathrm{Tri}(S, M)$ of ideal triangulations of $(S, M)$ forms a graph, where two triangulations are connected by an edge if they are related by a flip. (Since we have no punctures, there are no self-folded triangles.)

**Lemma 2.11.** *$\pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y_\phi)$ acts on the graph of ideal triangulations of $(S, M)$.*

*Proof.* Let $f \in \pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y_\phi)$. Fix an ideal triangulation $\Delta$, which defines a collection of matching spheres $\{L_e\}$ indexed by the edges of $\Delta$. We know that for each $e$ the Lagrangian sphere $f(L_e)$ is associated to some new matching path $\gamma_f(e)$. Moreover, Lemmas 2.8, 2.9 and 2.10 imply that $\gamma_f(e)$ and $\gamma_f(e')$ share exactly one respectively two end-points exactly when $e$ and $e'$ share exactly one respectively two end-points. The collection $\{\gamma_f(e)\}_{e \in \Delta}$ form a configuration of $4g - 4 + \sum k_i$ matching paths with no interior intersections, so these must form a maximal set of such arcs, hence are dual to the edges of some ideal triangulation. □

**Corollary 2.12.** *$\pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y_\phi)$ preserves $\mathrm{Stab}^\dagger(\mathcal{C}(S, M))$.*

*Proof.* It acts on the graph $\mathrm{Tri}(S, M)$, which is connected. This graph is identified [51] with the cluster exchange graph of reachable hearts. Therefore the group sends reachable hearts to reachable hearts, and hence preserves the distinguished component of the space of stability conditions. □

If we fix a generic ideal triangulation with no self-folded triangles, we get a finite collection of Lagrangian 3-spheres $L_e$ indexed by the dual 1-cycles to the edges $e$ of the triangulation, and these meet pairwise transversely. Let $Y_{\mathrm{pl}}$ denote the plumbing of copies of $T^*S^3$ according to the graph defined by the dual cellulation to the ideal triangulation. A Stein subdomain of the compact core of $Y_{\mathrm{pl}}$ embeds into $Y_\phi$, inducing an equivalence on compact and wrapped Fukaya categories. Let $T_e^*$ denote the dual cocore to $L_e$ (so $T_e^*$ is disjoint from $L_{e'}$ for $e' \neq e$, i.e., we are not

using Lefschetz thimbles but ascending manifolds of a suitable plurisubharmonic function, or cotangent fibres in a 'plumbing model').

**Lemma 2.13.** *In the previous situation:*

(1) *$Y_\phi$ admits a grading structure for which the algebra $\bigoplus_{e,e'} HF^*(L_e, L_{e'})$ is concentrated in non-negative degree and $\bigoplus_{e,e'} HW^*(T_e^*, T_{e'}^*)$ is concentrated in non-positive degree.*
(2) *The inclusion $\mathcal{C}(Q,W) \hookrightarrow \mathcal{F}(Y_\phi)$ is an equivalence, i.e., the compact cores generate the compact category. Thus $\mathcal{F}(Y_\phi)$ is equivalent to the dg-category of finite-dimensional modules over the Ginzburg algebra $\mathcal{J}(Q,W)$.*
(3) *The compact category $\mathcal{F}(Y_\phi)$ and wrapped category $\mathcal{W}(Y_\phi)$ are Koszul dual, and $\mathcal{W}(Y_\phi)$ is equivalent to the dg-category of all modules over $\mathcal{J}(Q,W)$.*

*Proof.* The first two results follow from [2]. Finite-dimensionality of

$$H^0(\mathcal{W}) = \bigoplus_{e,e'} HW^0(T_e^*, T_{e'}^*)$$

follows from the explicit computation in [43] using that around any puncture, the sum of the corresponding $d_e$ is strictly negative, because the underlying quadratic differential has a pole of order $d_i > 2$. Given that, the argument in [2, Proposition 4.2 and Lemma 4.6] shows that the compact cores generate the compact category. This argument crucially uses that, since the Floer algebra of the cotangent fibres is concentrated in non-positive degree, the ascending degree filtration on any module over $\bigoplus_{e,e'} HF^*(T_e^*, T_{e'}^*)$ is a filtration by $A_\infty$-submodules. We note [2, Lemma 4.12], which is a step towards [2, Proposition 4.2], uses a grading argument which is entirely local near each intersection point, providing one chooses gradings for all core components which are as symmetric as possible, cf. [2, equation (4.19)].

Koszul duality then follows from work of Kalck and Yang: finite-dimensionality of $H^0(\mathcal{W})$ (together with concentration of the whole algebra in non-positive degree) and [42, Proposition 2.5] shows that $HW^i(\mathcal{W})$ is finite-dimensional for each $i$, which fulfils the hypothesis of [42, Theorems 2.7 and 2.8]. The equivalence to the given Ginzburg algebras is established in [83] for the compact category and [17, 43] for the wrapped category. □

**Lemma 2.14.** *The Rabinowitz Fukaya category of $Y_\phi$ is the cluster category* $\mathrm{Clust}(Q,W)$ *of the quiver with potential $(Q,W)$. Moreover, the cluster tilting objects are in bijection with ideal triangulations.*

*Proof.* The first statement follows from the previous result, and the definition of the cluster category for a quiver with potential with finite-dimensional Jacobi algebra, cf. [4]. In the case of a marked bordered surface, the cluster-tilting objects are classified in [14]. □

**Lemma 2.15.** *There is a natural surjection* $\mathrm{Auteq}(\mathrm{Clust}(Q,W)) \longrightarrow \Gamma(S,M)$

*Proof.* We have an identification between the triangulation graph $\mathrm{Tri}(S,M)$ and the cluster exchange graph, so the given autoequivalence group acts by simplicial automorphisms of the triangulation graph. The group of such simplicial automorphisms was computed in [19] and is exactly $\Gamma(S,M)$. □

We now have a map

$$\mathrm{Sph}(S,M) \to \pi_0 \mathrm{Symp}_{\mathrm{ct}}(Y_\phi) \to \mathrm{Auteq}^\dagger(\mathcal{C}(Q,W))).$$

Our main result is completed on showing the following.

**Lemma 2.16.** *The composite map*

$$\pi_0 \operatorname{Symp}_{\mathrm{ct}}(Y_\phi) \to \operatorname{Auteq}^\dagger(\mathcal{C}(Q,W)) \to \Gamma(S,M)$$

*vanishes.*

*Proof.* The map $\operatorname{Auteq}^\dagger(\mathcal{C}(Q,W)) \to \Gamma(S,M)$ factors through autoequivalences of the cluster category, hence the composite map factors through the autoequivalence group of the Rabinowitz category, and so the result follows from Lemma 1.5. □

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